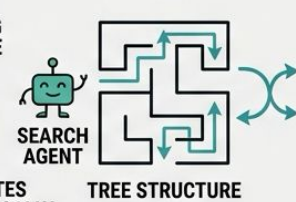




U.S. General Services  
Administration

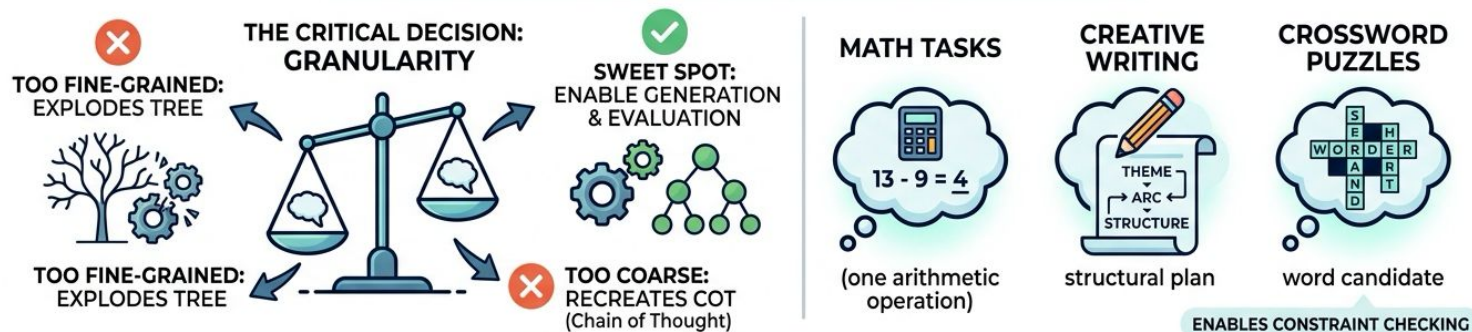
# Chapter 5.2: Tree-of-Thought (ToT) Fundamentals

US General Service Administration  
AI Community of Practice - March 2026



# Thought Decomposition & Generating Candidates

## OPTIMIZING THOUGHT DECOMPOSITION



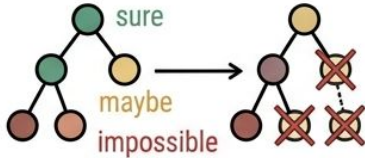
## GENERATING CANDIDATES: STRATEGIES & TRADE-OFFS



# Evaluating Branches -- Value Scores and Voting



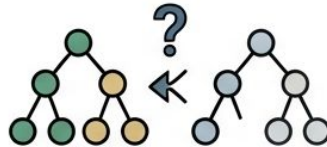
## Value estimation



Even noisy estimates  
dramatically reduce search space

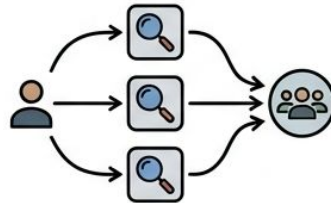


## Voting

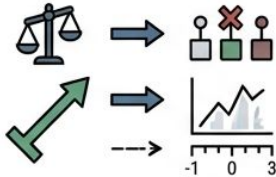


which is most promising?

## Ensemble sampling (3-5 times)



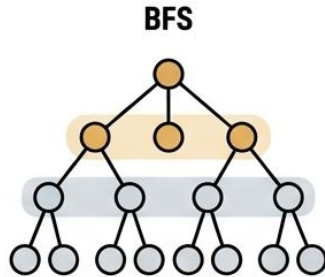
Improved evaluation reliability



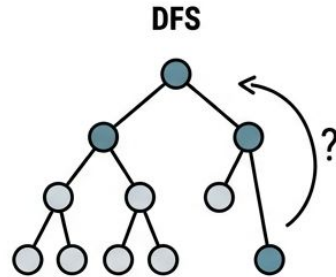
Comparative judgment is more  
reliable than absolute scoring

- Value estimation rates each state: "sure / maybe / impossible"
- Even noisy estimates dramatically reduce search space
- Voting compares candidates: "which is most promising?"
- Comparative judgment is more reliable than absolute scoring
- Ensemble sampling (3-5 times) improves evaluation reliability

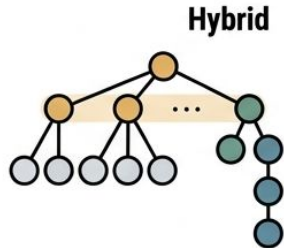
# Choosing a Search Strategy



Exploring breadth trees and kept at each depth level  
For shallow trees (2-4 steps) with uncertain evaluation



commits to the best branch, backtracks on failure  
Best for deep trees (5+ steps) with reliable evaluation



BFS at shallow depths, DFS for deep exploration

- BFS keeps top-b states at each depth level
- Best for shallow trees (2-4 steps) with uncertain evaluation
- DFS commits to the best branch, backtracks on failure
- Best for deep trees (5+ steps) with reliable evaluation
- Hybrid: BFS at shallow depths, DFS for deep exploration

# When NOT to Use ToT

**ToT IS NOT  
UNIVERSALLY SUPERIOR**



**CoT**

A SIMPLE,  
EFFECTIVE OPTION

**FACTUAL RECALL  
& LINEAR TASKS**



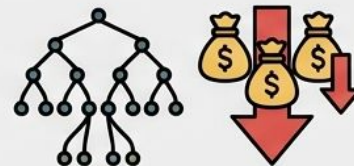
**NO GAIN FROM  
EXPLORATION**

**POOR  
DECOMPOSITION**



**MAKES ToT  
WORSE**

**AGGRESSIVE BACKTRACKING  
& TOKEN COST**

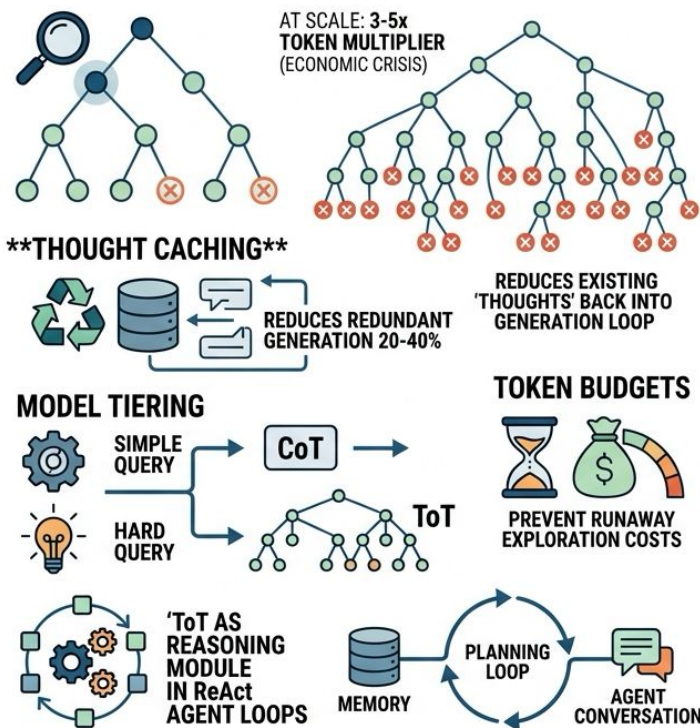


**TOKEN COST  
3-5x HIGHER THAN CoT**

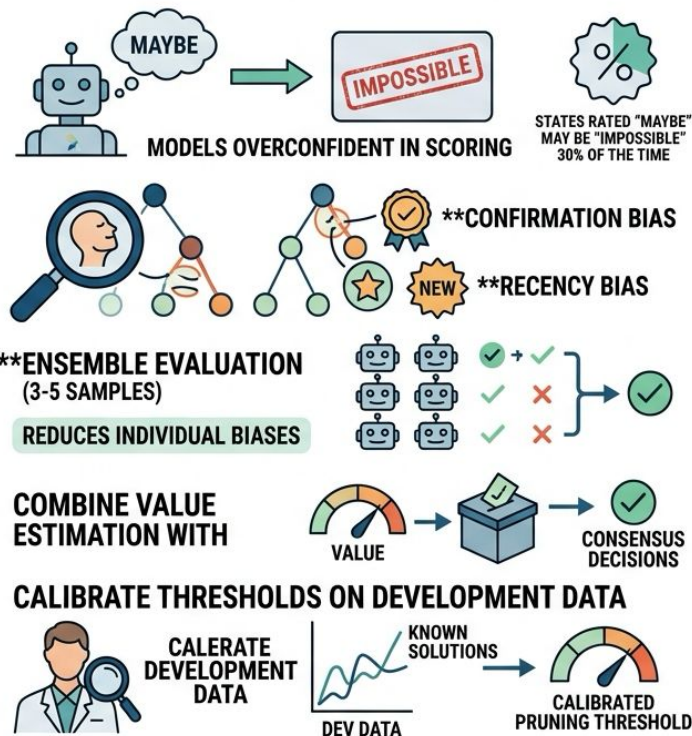


# Token Efficiency and Evaluation Calibration

## TOKEN EFFICIENCY & PRODUCTION INTEGRATION



## EVALUATION CALIBRATION: TRUST BUT VERIFY



# Real-World Applications of ToT

## REAL-WORLD APPLICATIONS OF TREE-OF-THOUGHT (ToT)



**MARKETING CONTENT**  
22% higher open rates  
via creative exploration



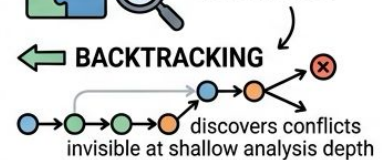
**ENTERPRISE ARCHITECTURE**  
prevents costly commitment  
to infeasible strategies



**CONFERENCE SCHEDULING**  
40-60 hours of manual  
work reduced to 8

### CONSTRAINT SATISFACTION

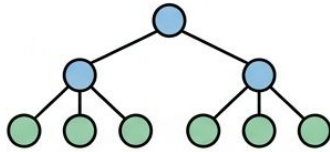
tasks with  
interdependencies  
benefit most



Looking ahead to discover details

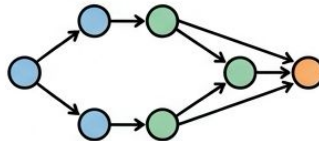
## EVOLVING REASONING: FROM TREES TO GRAPHS

### ToT (TREE-OF-THOUGHT)



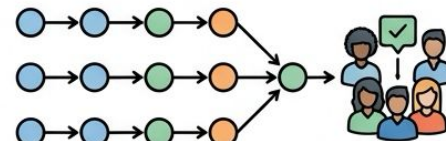
ToT maintains one parent per  
thought -- no reconvergence

### ADVANCING TO GRAPHS



Some problems need independent  
explorations to merge

### SELF-CONSISTENCY REASONING



Multiple reasoning paths  
vote on the final answer



**THE REASONING TOOLKIT**